

JASON PHILLIP LU

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EDUCATION

University of Michigan <i>Doctor of Philosophy, Civil Engineering</i> Advisor: Prof. Neda Masoud Specialization in Next Generation Transportation Systems	<i>Expected 2028</i> GPA: 3.94/4.0
University of Michigan <i>Master of Science, Industrial and Operations Engineering</i>	<i>2026</i> GPA: 3.94/4.0
Georgia Institute of Technology <i>Bachelor of Science, Industrial Engineering</i> Research Advisor: Prof. Pascal Van Hentenryck Minor in Scientific and Engineering Computing	<i>2022</i> GPA: 3.93/4.0 <i>Highest Honors</i>

RESEARCH INTERESTS

Methodologies optimization, machine learning, game theory, simulation, network modeling

Applications shared mobility, connected and autonomous vehicles, logistics, traffic control

PUBLICATIONS

*Corresponding Author

Peer-Reviewed Journals

Lu, J., Masoud, N. (2026). Incentive Design for Mobility Systems: A Review, Insights, and Future Directions. *Transportation Research Part E: Logistics and Transportation Review*. Volume 211, 104849. <https://doi.org/10.1016/j.tre.2026.104849>

Lu, J., Trasatti, A., Guan, H., Dalmeijer, K., Van Hentenryck, P. (2024). The Impact of Congestion and Dedicated Lanes on On-Demand Multimodal Transit Systems [Special Issue: Multimodal Public Transport, Travel Behaviour and Social Equity]. *Travel Behaviour and Society*. Volume 36, 100772. <https://doi.org/10.1016/j.tbs.2024.100772>

Under Review

Lu, J., Santanam, T., Guan, H., Riley, C., Kim, M.S., Trasatti, A., Masoud, N., Van Hentenryck, P. (2025). The Impact of Shared Autonomous Vehicles in Microtransit Systems: A Case Study in Atlanta. *Major Revision in Transportation Research Part C: Emerging Technologies*. <https://doi.org/10.48550/arXiv.2601.21046>

Lu, J., Xia, X., Bahrami, S., Masoud, N. (2026). Path-Based Incentives for Multimodal Networks under Multiclass Traffic Assignment. *Major Revision in Transportation Research Part B: Methodological*. <https://doi.org/10.13140/RG.2.2.15344.62722>

Lu, J., Masoud, N. (2026). The Incentive Bundle Program: Behaviorally-Consistent Incentive Design for Multimodal Transportation. *Submitted to Transportation Research Part B: Methodological*. <https://doi.org/10.13140/RG.2.2.22027.14882/1>

Working Papers

Lu, J., Masoud, N. Cooperation and Competition in Multimodal Transportation Incentive Programs: A Tri-level Game-Theoretic Optimization Approach. *Extended Abstract Accepted at the 2026 INFORMS Transportation Science and Logistics Conference.*

Lu, J., Chen, C., Liu, Y., Wu, Y.L., Kusari, A., Masoud, N. A Large-Scale Optimization and Simulation Framework for Sensor Configuration in Autonomous Vehicles. *Abstract Accepted at the 2026 INFORMS Annual Meeting.*

Xia, X., Masoud, N., **Lu, J.***. SignalAgentBench: A Reproducible Benchmark for Verifier-Gated LLM Agents in Traffic Signal Control. *Preprint Submitted to The 106th Transportation Research Board Annual Meeting.*

PRESENTATIONS

Invited Talks

A Large-Scale Optimization and Simulation Framework for Sensor Configuration in Connected and Autonomous Vehicles. *Next Generation Transportation Systems Seminar Series*, Department of Civil and Environmental Engineering, University of Michigan, March 19, 2026.

The Incentive Bundle Program to Enhance Multimodal Transport. *Next Generation Transportation Systems Seminar Series*, Department of Civil and Environmental Engineering, University of Michigan, February 27, 2025.

Conference Presentations

Lu, J., Chen, C., Liu, Y., Wu, Y.L., Kusari, A., Masoud, N. A Large-Scale Optimization and Simulation Framework for Sensor Configuration in Connected and Autonomous Vehicles. *2026 INFORMS Annual Meeting*, San Francisco, California, November 1-4, 2026.

Lu, J., Masoud, N. Cooperation and Competition in Multimodal Transportation Incentive Programs: A Tri-level Game-Theoretic Optimization Approach. *2026 INFORMS Transportation Science and Logistics Conference*, Boston, Massachusetts, July 26-29, 2026.

Masoud, S., Masoud, N., Alinezhad, E., **Lu, J. (presenter)**, Rapp, S., Rimanelli, J. Enhancing Military Digital Twins: Leveraging Dynamic Data-Driven Application Systems for Complex Operational Scenarios. *2025 ARC Annual Program Review*, Ann Arbor, Michigan, June 17-18, 2025.

Lu, J., Chen, C., Liu, Y., Wu, Y.L., Kusari, A., Masoud, N. A Large-Scale Optimization and Simulation Framework for Sensor Configuration in Connected and Autonomous Vehicles. *2026 CCAT Global Symposium on Mobility Innovation*, Ann Arbor, Michigan, April 17, 2026.

Masoud, N., **Lu, J.**, Chen, C., Liu, Y., Wu, Y., Kusari, A. Balancing Safety, Cybersecurity, and Mobility: Quantifying the Impact of Sensor Redundancy in Connected and Autonomous Vehicles. *2026 CCAT Global Symposium on Mobility Innovation*, Ann Arbor, Michigan, April 17, 2026.

Lu, J., Xia, X., Bahrami, S., Masoud, N. Can Incentives Reshape How We Move? A Multimodal Traffic Assignment Perspective. *The 105th Transportation Research Board Annual Meeting*, Washington, D.C., January 11-15, 2026.

Lu, J., Masoud, N. A Game Between Ridehailing Operator and Government Under the Incentive Bundle Program. *2025 INFORMS Annual Meeting*, Atlanta, Georgia, October 26-29, 2025.

Lu, J., Masoud, N. The Incentive Bundle Program to Enhance Multimodal Transportation. *The 7th Bridging Transportation Researchers Conference*, Virtual, August 6-7, 2025.

Masoud, S., Masoud, N., Alinezhad, E., **Lu, J. (presenter)**, Rapp, S., Rimanelli, J. Adaptive Contested Logistics: A Dynamic Data-Driven Application System Framework with Dynamic

Bayesian Belief Networks. *2025 ARC Annual Program Review*, Ann Arbor, Michigan, June 17-18, 2025.

Masoud, N., **Lu, J.**, Chen, C., Liu, Y., Wu, Y., Kusari, A. Balancing Safety, Cybersecurity, and Mobility: Quantifying the Impact of Sensor Redundancy in Connected and Autonomous Vehicles. *2025 CCAT Global Symposium on Mobility Innovation*, Ann Arbor, Michigan, March 28, 2025.

Lu, J., Masoud, N. Designing Incentive Bundles for Multimodal Transportation Systems. *2024 INFORMS Annual Meeting*, Seattle, Washington, October 20-23, 2024.

Akhlaghi, V.E., Guan, H., **Lu, J.**, Van Hentenryck, P. Optimizing Truck Fleet Scheduling for Fuel Deliveries. *2023 INFORMS Annual Meeting*, Phoenix, Arizona, October 15-18, 2023.

ADVISING

Masters Students

Xingpeng Xia, University of Michigan
M.S. Civil Engineering (Expected 2026)

Yuyang Liu, University of Michigan
M.S. Electrical Engineering (2026)
Placement: PhD Student, University of Michigan

Undergraduate Students

Tianlang Chen, University of Michigan
B.S. Data Science (Expected 2027)

Yi Ling Wu, University of Michigan
B.S. Computer Engineering (2025)
Placement: PhD Student, Texas A&M University
– Aviles-Johnson Doctoral Fellowship

TEACHING

Graduate Student Instructor

CEE 553 - Infrastructure Systems Optimization
Instructor: Prof. Neda Masoud

University of Michigan

Fall 2026

Guest Lecturer

Gurobi
CEE 553 - Infrastructure Systems Optimization

University of Michigan

Fall 2024, Fall 2025

Overview of Next Generation Transportation Systems Program
CEE 200 - Introduction to Civil and Environmental Engineering

Spring 2025

Preparing for the Civil and Environmental Engineering Career Fair
CEE 200 - Introduction to Civil and Environmental Engineering

Fall 2024

Head Teaching Assistant

Georgia Institute of Technology

ISYE 4134 - Constraint Programming
Instructors: Mr. Tejas Santanam, Prof. Pascal Van Hentenryck

Spring 2023

Undergraduate Teaching Assistant

Georgia Institute of Technology

ISYE 3044 - Simulation Analysis and Design
Instructor: Prof. Seong-Hee Kim

Summer 2022

ISYE 4034 - Decision and Data Analytics

Spring 2022

Instructor: Prof. Jye-Chyi Lu

ISYE 3770 - Statistics and Applications

Spring 2021

Instructor: Dr. Tuba Ketenci

HONORS AND AWARDS

Honorable Mention, Student Poster Competition

2025

2026 INFORMS Transportation Science and Logistics Conference

Western Track Best Paper Award

2025

The 7th Bridging Transportation Researchers Conference

Rackham Conference Travel Grant

2024, 2025

Rackham Graduate School, University of Michigan

Seth Bonder Fellowship

2022, 2023

H. Milton Stewart School of Industrial and Systems Engineering, Georgia Institute of Technology

Stewart Topper Fellowship (Declined)

2023

H. Milton Stewart School of Industrial and Systems Engineering, Georgia Institute of Technology

Senior Design Capstone Finalist

2022

H. Milton Stewart School of Industrial and Systems Engineering, Georgia Institute of Technology

SERVICE - INTERNAL

Michigan Transportation Student Organization

Vice President

May 2024 - Apr. 2025

The Seth Bonder Camp in Computational and Data Science for Engineering

Instructor

Jun. 2022 - Jul. 2023

SERVICE - EXTERNAL

Reviewer

Peer-Reviewed Journals

- Transportation Research Part B: Methodological

Conferences

- The 105th Transportation Research Board Annual Meeting
- The 7th Bridging Transportation Researchers Conference
- The 104th Transportation Research Board Annual Meeting

Session Chair

- Advancements in Urban Transit Systems, *2024 INFORMS Annual Meeting*, Seattle, Washington, October 20-23, 2024.

INDUSTRY EXPERIENCE

Convoy

Process Improvement Intern

Jan. 2022 - May 2022

Yokogawa

Industrial Engineering Intern

May 2021 - Dec. 2021

Industrial Engineering Intern

May 2020 - Jul. 2020